

A Study on
“Inflation In India”

A Project Submitted for the Degree of
MASTERS IN BUSINESS ADMINISTRATION

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DECLARATION

We Hereby declare that the project entitled “**Inflation in India**” submitted to the fulfilment for award of the **MASTER OF BUSINESS ADMINISTRATION (MBA) DEGREE AT CENTURION UNIVERSITY OF TECHNOLOGY AND MANAGEMENT**. This Project works is original and has not been submitted so far in any part or full for any other university or institute for the award of any Degree or Diploma

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BONAFIDE CERTIFICATE

Certified that this project report “**INFLATION IN INDIA**” is the Bonafide work of “V. Venkata Raj Gopal, K. Sree Harini, M. Krishna Padma, CH. Sravani” who carried out the project work under my supervision. This is to further certify, to the best of my knowledge, that this project has not been carried out earlier in this institute and the university.

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INFLATION IN INDIA

CHAPTER 1

1.1 Introduction

Inflation is one of the most important concepts in economics because it affects the day-to-day life of every citizen. Inflation refers to a sustained rise in the general price level of goods and services over a period of time. When inflation increases, the purchasing power of money decreases. This means that a person must spend more money to buy the same product or service compared to earlier. Inflation impacts households, businesses, investors, and the government, making it a key indicator of economic health.

In India, inflation is mainly measured through the Consumer Price Index (CPI) and the Wholesale Price Index (WPI). CPI measures the change in prices at the consumer level, while WPI tracks prices at the wholesale or producer level. Additionally, India also looks at core inflation, which excludes food and fuel prices because these two categories are highly volatile. The Reserve Bank of India (RBI) is responsible for controlling inflation using monetary policy tools such as the repo rate, reverse repo rate, and open market operations. India follows an inflation-targeting framework where the ideal inflation rate is 4 percent with an allowance of ± 2 percent, meaning inflation should remain between 2 percent and 6 percent.

The period between 2020 and 2025 has been a highly volatile time for inflation in India. In 2020, the COVID-19 pandemic disrupted supply chains, reduced production, and restricted the movement of goods, causing shortages and rising prices. In 2021, the economy reopened, but inflation remained high due to global shipping delays and rising demand. In 2022, the Russia–Ukraine conflict caused fuel, wheat, edible oil, and fertilizer prices to rise sharply across the world, including India. This further pushed inflation to uncomfortable levels.

Inflation affects various sectors differently. Food inflation impacts households the most, especially lower-income families. Fuel inflation influences transportation and logistics costs, which then increases the prices of all other goods. High inflation also impacts savings, investments, interest rates, and business costs. This makes inflation control essential for maintaining economic stability.

Therefore, studying inflation during the 2020–2025 period is important because of the multiple global and domestic shocks that influenced price levels. Understanding inflation helps policymakers design effective strategies to reduce price volatility, protect consumers, and support long-term economic growth. This study aims to provide a clear analysis of inflation trends, causes, and consequences in India during this period.

Inflation is not only an economic concept but also a powerful social and political factor that shapes the conditions of a country. When inflation rises for a long period, it can reduce the affordability of essential goods like food, medicines, housing, transportation, and education. For developing nations like India, where a large percentage of the population belongs to middle- and lower-income groups, even a small rise in inflation can create financial stress. Inflation also affects consumer confidence. When people believe that prices will continue to rise, they tend to increase their spending immediately, which further adds pressure on demand. In contrast, when inflation is low and stable, people plan their budgets better, save more, and invest with confidence.

Inflation is also influenced by both demand-side and supply-side factors. Demand-pull inflation happens when the demand for goods and services increases faster than supply, often during economic recovery periods. Supply-side inflation, also known as cost-push inflation, happens when production costs rise. This includes higher fuel prices, increased wages, expensive raw materials, or disruptions in supply chains. During 2020–2025, India experienced both types of

inflation. For example, supply shocks due to lockdowns created shortages (cost-push), while post-pandemic recovery increased demand (demand-pull).

Another important aspect of inflation is its relationship with employment and economic growth. Moderate inflation is usually associated with economic expansion because businesses produce more, hire more workers, and invest more. However, when inflation becomes too high, businesses struggle with higher costs and reduced consumer demand. This can slow down growth and negatively impact job creation. Therefore, maintaining the right balance between inflation and growth is essential.

Inflation also influences the currency value. When inflation rises faster in India compared to other countries, the value of the Indian rupee may fall against global currencies like the US dollar. A weaker rupee can make imports more expensive, leading to further inflation, especially in sectors dependent on foreign goods such as petroleum, technology, and machinery. This shows how inflation is deeply connected to international trade and global economic conditions.

In the Indian context, inflation also has implications for poverty and inequality. Rising prices reduce the real value of wages and savings, hitting poorer households harder because they spend a larger share of their income on essentials like food and fuel. Persistent inflation can widen economic inequality and reduce overall well-being.

1.2 Background of the Study

The background of this study focuses on understanding how inflation behaved in India during the years 2020 to 2025, a period marked by unprecedented global and domestic economic challenges. Inflation is not only affected by internal factors like production, agriculture, and demand but also by external events such as global oil prices, international conflicts, global supply chain disruptions, and currency fluctuations. The chosen period represents one of the most economically unpredictable times in recent history.

In 2020, the COVID-19 pandemic hit the world, leading to nationwide lockdowns in India. Movement restrictions affected transportation, supply chains, and labor availability. Shortages of essential goods led to a rise in prices, especially for vegetables, pulses, edible oils, and medical items. Many small businesses shut down temporarily, which reduced supply further. Demand for certain goods increased, leading to price pressure in multiple sectors.

In 2021, as the economy began to reopen, inflation did not reduce as expected. Instead, global shipping delays, shortages of raw materials, and high freight charges kept prices elevated. Edible oil prices rose sharply due to limited global supply. The second wave of COVID-19 in India also disrupted supply chains again.

In 2022, inflation worsened due to the Russia–Ukraine war. The conflict affected global crude oil prices, natural gas supply, wheat exports, fertilizer availability, and edible oil production. India, being heavily dependent on crude oil imports, felt the impact immediately. Transportation and manufacturing costs increased, leading to broad inflation across sectors. Food inflation also remained high due to global shortages.

In 2023 and 2024, the Reserve Bank of India increased interest rates several times to control inflation. Although these measures helped reduce inflation gradually, food and fuel prices continued to fluctuate due to climate changes, irregular monsoons, global oil movements, and rupee depreciation.

By 2025, inflation trends remained influenced by global uncertainty, oil markets, and domestic supply conditions. This period highlights how vulnerable India is to both domestic challenges and international events.

1.3 Need for the Study

The need for this study arises from the fact that inflation directly influences the life of every individual and plays a central role in shaping the economic stability of a country. Understanding inflation is essential for policymakers, businesses, households, financial institutions, and researchers. The period from **2020 to 2025** has been particularly unique due to major global disturbances such as the COVID-19 pandemic and the Russia–Ukraine war. These events significantly impacted inflation levels across the world, including India.

Firstly, inflation affects the **purchasing power** of households. When prices of essential goods like food, vegetables, milk, and fuel increase, families—especially middle- and lower-income groups—struggle to manage their budgets. This reduces their real income and their ability to save for the future. Therefore, studying inflation helps understand how rising prices impact the standard of living.

Secondly, inflation influences **business operations**. When the cost of raw materials, fuel, transportation, and labor increases, companies face higher production costs. Businesses may increase prices to maintain profits, which further adds to inflationary pressure. Understanding inflation helps businesses in planning production, pricing, and inventory decisions.

Thirdly, inflation is crucial for **government policy-making**. The government frames the Union Budget, subsidies, welfare schemes, and taxation policies based on inflation levels. High inflation requires the government to step in with price control measures such as export restrictions, increased imports, and subsidy adjustments. Without detailed inflation analysis, policymaking becomes inefficient.

Fourthly, inflation affects the **financial sector**. The Reserve Bank of India uses tools like repo rates and monetary tightening to manage inflation. A detailed study helps understand how these policy actions impact borrowing costs, bank interest rates, investments, and liquidity in the economy.

Thus, this study is essential for understanding how inflation behaved during a period of high uncertainty and how India responded through policy actions and economic adjustments

1.4 Objectives of the Study

The main objective of this study is to analyze and understand the inflation trends in India during the period **2020 to 2025**. Inflation during these years shows significant variation due to pandemic-related disruptions, fuel price shocks, global conflicts, and domestic supply challenges. Therefore, clearly defined objectives help guide the research in a focused and meaningful direction.

Primary Objective

- To analyze the inflation trends in India from 2020 to 2025.

This primary objective aims to provide a clear understanding of how inflation changed year by year and what factors contributed to these changes.

Secondary Objectives

1. To study the factors influencing inflation in India.

Inflation is shaped by several elements such as fuel prices, food supply, international trade, currency movements, and government policies. Understanding these factors helps identify the root causes of price fluctuations.

2. To analyse food inflation during the study period.

Food items contribute the largest share to the Consumer Price Index (CPI). During 2020–2025, food inflation was highly unstable due to supply issues, climatic changes, and global shortages. Studying food inflation helps understand its impact on consumers.

3. To examine fuel inflation trends.

Fuel prices significantly affect transportation, logistics, and production costs. India depends heavily on imported crude oil, making its inflation sensitive to global oil markets. This objective examines the role of fuel in overall inflation.

1.5 Scope of the Study

The scope of this study defines the areas covered, the time period considered, the data used, and the boundaries within which the research is conducted. This study focuses on the analysis of inflation trends in India from **2020 to 2025**, a period selected due to highly unusual and impactful events that shaped the Indian economy.

Geographical Scope

The study is limited to **India**, focusing on nationwide inflation trends. It does not include state-wise inflation comparisons unless required for explanation. National-level data is used because it provides a broad understanding of price changes affecting the entire country.

Time Scope

The study covers a six-year period from **2020 to 2025**. These years include critical events such as the COVID-19 pandemic, economic recovery phases, the Russia–Ukraine war, supply chain disruptions, and RBI's monetary policy interventions. This timeframe helps capture short-term and long-term inflation patterns.

Data Scope

The study relies entirely on **secondary data** from the following sources:

- Consumer Price Index (CPI)
- Ministry of Statistics and Programme Implementation (MOSPI)

1.6 Limitations of the Study

Every research study has certain limitations that define what it cannot fully cover or control. The present study on inflation in India from **2020 to 2025** also includes some limitations, mainly due to data availability, time constraints, and the nature of secondary information.

4. Dependence on Secondary Data

This study is entirely based on secondary data collected from government reports, websites, journals, and statistical databases. Secondary data may sometimes contain delays, revisions, or

updates that affect accuracy. The study does not use primary data such as surveys or interviews, which limits firsthand insights.

2. Rapidly Changing Economic Conditions

Inflation is affected by global and domestic events which change frequently. For example, oil prices, monsoon conditions, global conflicts, and policy announcements can shift inflation trends instantly. Because of this dynamic nature, the conclusions of the study may change with new data releases.

3. Limited Access to Real-Time Data

Government departments such as MOSPI and RBI release inflation data periodically, often with revisions. Not all data for the year 2025 may be fully available at the time of research, which limits complete year-wise analysis.

4. Limited Coverage of All Influencing Factors

Inflation is influenced by hundreds of factors such as consumer behaviour, agricultural yield, industrial growth, global commodity prices, and political decisions. It is not possible to include all these variables in the study. Only major factors like food, fuel, supply chains, and global events are analysed.

CHAPTER 2

REVIEW OF LITERATURE

Ball, Chari and Mishra (2016) studied the behaviour of inflation in India and explained how general inflation, food inflation and fuel-related inflation together influenced overall price levels in the country. The authors analysed India's inflation patterns from the mid-1990s and found that headline inflation is much more volatile than core inflation because it is heavily affected by sudden changes in food and fuel prices. Their study showed that food inflation, especially in items like pulses, milk, vegetables and fruits, frequently caused large price spikes that pushed up overall inflation. Fuel and energy prices also played a major role, as increases in crude oil, coal, electricity and petroleum products created upward pressure on transportation and production costs, which ultimately raised the general inflation rate. They explained that these supply shocks—mostly coming from food and fuel—create temporary but strong jumps in headline inflation, and these shocks sometimes influence inflation expectations for the future. The authors also found that India's inflation often behaves like advanced economies of the 1970s and 1980s, where price setters adjusted prices based on past inflation, causing inflation to persist for longer periods. Their analysis further showed that general inflation in India is influenced not only by current food and fuel prices but also by the level of economic activity, meaning that when output rises above its normal level, inflation tends to increase. They used a Phillips Curve framework and concluded that controlling inflation in India requires understanding both demand-side effects and supply-side pressures, particularly those coming from agriculture and energy. Another key finding was that food and fuel inflation contribute significantly to the “right tail” of price distributions, meaning they are responsible for extreme price increases that push general inflation upward. The study highlighted that nearly two-thirds of these extreme inflation spikes are due to food and energy prices alone. This shows that India's inflation is highly sensitive to supply constraints, global fuel price changes, and structural issues in agriculture. They emphasised that while general inflation can be managed through monetary policy, food and fuel inflation require broader reforms in supply chains, agricultural markets, storage systems, and energy policies. Overall, their work provides strong evidence that food inflation and fuel & light inflation are the main drivers of India's overall inflation trends and must be carefully monitored for effective inflation control.

Mohanty & John (2014) studied what drives inflation in India using quarterly data from 1996–97 to 2013–14 and a time-varying SVAR model. They focused on overall (general) inflation measured by WPI and examined key contributors like crude oil, the output gap, fiscal deficit and monetary policy. The paper finds that crude oil (fuel & energy) was a dominant driver of inflation during 2009–2011, pushing up transport and production costs. Food price movements also show up strongly in WPI fluctuations, causing many of the large, short-term jumps in headline inflation. Monetary policy's influence on inflation stayed roughly steady over the period, suggesting RBI actions mattered but did not fully explain large swings. The output gap affected inflation asymmetrically and its influence weakened in recent years, making demand-driven inflation less clear-cut. Fiscal deficits became an important contributor around 2011–2012, adding to inflationary pressures. The authors also show that inflation dynamics changed over time, so drivers that mattered in one sub-period might be weaker or stronger later. Overall, the study highlights that fuel & light (oil/energy) and food supply shocks explain many extreme episodes of general inflation, while policy and output effects vary with time.

Dua & Goel (2021) examined the key drivers of overall inflation and food inflation in India using monthly data from 1996 to 2017. Their study showed that inflation in India has strong long-run relationships with factors such as expected inflation, output gap, money supply growth, exchange rate, interest rates, fiscal deficit, rainfall, and international oil and food prices. Among these, expected inflation, exchange rate movements, and money supply growth were found to be the most powerful determinants of general inflation. The authors also highlighted that global fuel prices and international food prices played a major role in pushing both general inflation and food inflation upward, indicating India's high dependence on external supply conditions. The study further found that shocks to food and fuel inflation strongly influence people's inflation expectations, which then spread into overall inflation and make price pressures persist for longer periods. Their results suggest that while monetary policy can influence demand-driven inflation, it has limited control over food inflation and fuel & light inflation, since these are mostly supply-driven. Thus, inflation management becomes more challenging in an economy like India where external shocks, agricultural dependence, and fuel imports play a major role. Overall, the authors conclude that understanding the combined impact of food prices, fuel prices, global conditions, and domestic demand factors is essential for predicting and managing India's inflation effectively.

Callen and Chang (1999) studied how inflation in India behaves and which indicators help predict future price trends. They found that even though India moved away from a strict money-supply targeting system, monetary aggregates like M1 and M3 still give strong signals about future general inflation. Their analysis showed that rises in money supply, import prices, and exchange rate depreciation often lead to higher inflation, especially in sectors linked to manufacturing and energy. The authors observed that fuel-related factors such as import prices and the rupee-dollar exchange rate are important early indicators of inflation pressure. They also noted that food prices, driven by rainfall and agricultural supply shocks, create large fluctuations in primary product inflation, which then spills over into general inflation. The paper concluded that while monetary policy can help manage overall demand, supply-side factors—especially fuel price movements and food supply variations—play a major role in pushing inflation up in India. Therefore, the RBI must monitor a wide set of indicators, not just money growth, to forecast inflation effectively.

Bhattacharya and Sen Gupta (2015) studied why food inflation in India remained extremely high between 2006 and 2013 and found that both demand and supply pressures were responsible. They showed that rising incomes led people to shift their diets toward milk, meat, fish, fruits and vegetables, but the supply of these items did not grow fast enough, creating persistent excess demand and pushing up food prices. On the supply side, the authors found that agricultural wages grew sharply after MGNREGA, and this rise in labour cost became one of the strongest drivers of food inflation across almost all commodities. Fuel inflation had a moderate but clear impact because higher fuel prices raised transportation costs and agricultural input costs, influencing farmgate to retail price spreads. Minimum Support Price (MSP) hikes were also identified as a major contributor, especially for cereals, pulses and sugar, because higher procurement prices pushed wholesale prices upward. Global food prices affected India only in tradeable commodities like edible oil and sugar. The study also showed that food inflation has strong pass-through effects on general inflation, raising both non-food inflation and overall inflation due to wage pressures and household expectations. Overall, the authors concluded that India's high food inflation was driven by a combination of rising demand, structural supply constraints, rising input costs, MSP policies, and moderate fuel-related cost pressures, making food inflation a major contributor to general inflation in the economy.

Bhattacharya and Sen Gupta (2017) analysed why India experienced very high food inflation between 2006 and 2013 and found that the rise was driven mainly by strong domestic cost pressures. Their study shows that agricultural wage inflation became the most important and consistent driver of food inflation, especially after the expansion of MGNREGA, which pushed rural wages upward and increased the cost of production. They found that fuel inflation had a moderate but noticeable effect, because higher fuel prices raised transportation and input costs, although the impact varied across commodities. The authors also observed that global food prices played only a limited role, except for tradeable items like edible oils and sugar. Their SVAR and SVECM results showed that food inflation has strong spillover effects on both non-food inflation and overall general inflation, meaning that rising food prices quickly spread into the entire economy. They concluded that food inflation in India is shaped mainly by domestic structural factors—rising wages, supply shortages, and cost pressures—while external factors matter only for a few commodities. The study highlights that controlling food inflation is crucial because it has a direct and persistent impact on headline inflation and household living costs.

Chand et al. (2011) examine the sharp rise in India's food inflation since 2008 and diagnose it as a product of interacting domestic supply shocks, distributional inefficiencies, and demand-side liquidity — not just a one-off global price shock. They show food inflation shifted from grains and processed foods (2008) to high-value items (milk, vegetables, meat) by 2010, with episodic spikes (onion) caused by localized production losses and weak post-harvest logistics. Key supply-side problems are stagnating area/yields, poor value-chain infrastructure, high post-harvest losses, fragmented mandis and hoarding. Demand-side contributors include large fiscal and monetary stimulus, rising rural incomes (MGNREGA, pay commissions), and rapid M3 growth that amplified pressure on food prices — especially on high-elasticity items. The brief stresses that short-run fixes (drawdown of public stocks, targeted imports, liquidity withdrawal) must be paired with structural reforms: mandi/marketing reforms, farm–firm linkages, private investment in cold chains/storage, better market intelligence, and targeted investments in backend services. Policy recommendations emphasize coordinated, phased actions — combine immediate supply/market measures with long-term reforms in infrastructure, marketing institutions and agricultural incentives to sustainably tame food inflation. (Policy Brief, NCAP/IFPRI, Feb 2011).

Moorthy & Kolhar (2011) analyse how rising food prices create major challenges for India's monetary policy, especially in an economy where poor households spend a large share of their income on food. Using both historical evidence and a two-commodity economic model, the authors show that food-price shocks behave very differently from normal demand-driven inflation. Their model demonstrates that when food prices rise, the CPI increases sharply, but the GDP deflator may actually fall, creating a misleading impression that overall inflation is lower than what consumers experience. The study argues that supply shocks—especially abnormal rainfall, poor weather, and structural bottlenecks—must be analysed commodity-by-commodity rather than treated as temporary disturbances. They also show that high food inflation can reduce welfare even during periods of strong GDP growth, with the poor suffering the most. The paper concludes that India's monetary policy framework should shift its focus toward a population-weighted CPI, because it better reflects the true inflation faced by households compared to the GDP deflator. The authors warn that ignoring food inflation can lead to policy mistakes, as food-price shocks create long-lasting effects on inflation expectations, consumption patterns, and welfare.

Patnaik (2021) investigates whether food inflation in India creates strong “second-round effects,” meaning that persistent food price increases spill over into headline inflation and eventually push up core inflation as well. Using frequency-domain Granger causality methods, the study finds that food inflation directly causes headline inflation (first-round effect), and headline inflation significantly causes core inflation (second-round effect), showing that food shocks do not fade quickly in India. The paper challenges the traditional belief that food inflation is temporary and too volatile to influence underlying inflation trends; instead, results show that food inflation in India is actually **less volatile** than expected and has long-lasting consequences. The study also shows that households consistently **overestimate future inflation**, with expectations 3–4 percentage points above actual inflation, mainly because food price shocks heavily influence people's perception of future prices. This weak anchoring of inflation expectations reduces the effectiveness and credibility of monetary policy. Further, Patnaik finds that monetary policy influences food inflation only in the short and medium run, but has very limited effect on headline and core inflation in the short term. The paper concludes that managing food inflation and anchoring expectations must be central to India's inflation-targeting framework because

uncontained food shocks can permanently push up overall inflation and weaken policy transmission.

Ghosh (2022) studied how India's diesel subsidy reforms (2012–2014) affected inflation and economic growth. Since India imports most of its oil, diesel prices play a big role in transport costs and overall inflation. Earlier, diesel was heavily subsidized, which kept transport costs low but increased the government's fiscal burden. The government removed these subsidies and started linking diesel prices to the market, but only **partially passed on** the fall in global oil prices to consumers. Using a macroeconomic model with data from 1997 to 2017, the study finds that diesel subsidy reform **helped reduce inflation**, improved government finances, and gave stability to the economy during a time when exports and rainfall were weak. However, economic growth did not rise as much as expected because other growth factors—like exports, government spending, and money supply—were slow after 2012. The study also shows that if India had adopted a **fully market-linked diesel price**, the economy could have grown even faster. Overall, the paper concludes that diesel subsidy reforms were beneficial, but better results would have come from passing the full global price reduction to consumers.

CHAPTER 3

3.1 Meaning and Concepts of Inflation

Inflation is one of the most important concepts in economics because it affects every individual, household, business and government decision. In simple words, inflation means that the overall price of goods and services is rising continuously over time, and as a result, the value of money decreases. For example, if a litre of milk cost ₹40 last year but costs ₹50 this year, there is inflation. When inflation rises, people need to spend more money to buy the same quantity of goods, which directly affects their standard of living.

Inflation does not happen suddenly; it occurs due to several reasons. One major reason is increased demand, where more people want to buy goods, but supply cannot keep up. This is called demand-driven inflation. Another reason is increased production costs, such as higher fuel prices, wage increases, or scarcity of raw materials. When businesses face higher costs, they increase prices, causing cost-push inflation. Inflation can also rise when the money supply in the economy increases too quickly. With more money in circulation, people spend more, which pushes prices upward.

Inflation affects different groups in different ways. Salaried individuals and pensioners may suffer because their income may not increase at the same speed as prices. Borrowers may benefit because the real value of their debt reduces over time. Businesses face uncertainty because rising prices make it difficult to plan production and investment. On a national level, high inflation reduces purchasing power, increases inequality, discourages savings, and creates instability in the economy.

In India, inflation is closely monitored through indicators like the Consumer Price Index (CPI). The Reserve Bank of India (RBI) controls inflation by adjusting interest rates and managing liquidity in the financial system. When inflation rises too much, RBI increases interest rates to reduce spending. When inflation is low and the economy needs support, RBI may lower interest rates to encourage borrowing and investment.

Overall, inflation is not always harmful. A moderate level of inflation is considered normal for a growing economy, as it encourages spending and investment. However, high or unpredictable inflation is dangerous because it creates uncertainty and reduces economic growth. Understanding the meaning and concepts of inflation is important for analysing how price changes impact people's daily lives and how policies can help manage inflation effectively.

3.2 Types of Inflation

Inflation does not occur in the same way every time. It has multiple forms depending on the cause, intensity, and speed at which prices increase. Understanding the types of inflation helps policymakers and researchers analyse the reasons behind rising prices and design appropriate solutions.

One common type is **Demand-Pull Inflation**, which happens when the total demand for goods and services exceeds the available supply. In a fast-growing economy, when incomes rise and consumers spend more, demand increases sharply. If industries cannot produce goods fast enough to match this rising demand, prices begin to rise. This type of inflation often occurs during periods of economic growth.

Another type is **Cost-Push Inflation**, which arises when the cost of producing goods increases. If wages, fuel prices, transportation costs, or raw material prices go up, producers pass these higher costs on to consumers by increasing prices. Cost-push inflation has been very common in India due to fuel price hikes, supply shortages, and disruptions in agricultural supply chains.

Built-In Inflation refers to the inflation caused by expectations. When people believe prices will rise in the future, they adjust their behaviour—workers demand higher wages, businesses raise prices in advance, and consumers buy more before prices increase. This creates a cycle where inflation fuels more inflation.

Food Inflation is extremely important in India because food has a high weight in the consumption basket. When prices of essential items like vegetables, cereals, pulses, milk, or edible oils increase, food inflation rises. This type of inflation is usually caused by poor monsoons, crop failures, storage issues, or supply chain disruptions. Since low-income households spend a large portion of their income on food, food inflation hits them the hardest.

Fuel and Light Inflation refers to price increases in petrol, diesel, LPG, kerosene, and electricity. Fuel inflation has a major indirect impact. When fuel becomes expensive, transportation and production costs rise, leading to higher prices for almost all goods and services.

Inflation can also be classified based on intensity, such as creeping inflation (slow), walking inflation (moderate), galloping inflation (fast), and hyperinflation (extremely high).

Understanding these types is crucial because each kind of inflation requires a different policy response. For example, demand-pull inflation may require monetary tightening, while cost-push inflation may need supply-side measures. In India's recent years, food and fuel inflation have played major roles in shaping headline inflation, making these categories essential for study.

3.3 Measurement of Inflation (CPI)

Inflation can only be controlled and analysed properly if it is measured accurately. In India, the most widely used measure of inflation is the **Consumer Price Index (CPI)**. CPI tracks the change in retail prices of a basket of goods and services that households commonly purchase. It is considered the best representation of the cost of living because it reflects how much consumers actually spend on essential items.

CPI includes a wide range of consumption categories such as food and beverages, fuel and light, housing, clothing, footwear, transport, education, and medical services. Each category is assigned a **weight** based on its importance in the average household budget. Since food is a major part of consumer spending in India, it has the highest weight in CPI. This means changes in food prices significantly impact overall inflation.

The CPI Inflation rate is calculated by comparing the current month's CPI value with the value from the same month of the previous year. This helps us understand how much prices have increased over a year.

$$\text{Inflation Rate} = (\text{CPI this year} - \text{CPI last year}) / \text{CPI last year} \times 100$$

There are different types of CPI used in India:

- **CPI for Rural (CPI-R)** – reflects price changes in rural households
- **CPI for Urban (CPI-U)** – reflects price changes in urban households
- **CPI Combined (CPI-C)** – the official inflation measure used by RBI

The CPI-C is crucial for policymaking. Since India adopted an inflation-targeting framework, the RBI aims to maintain inflation at around 4% with a tolerance band of $\pm 2\%$. CPI helps RBI understand whether inflation is rising due to food prices, fuel costs, or other factors, allowing them to adjust interest rates accordingly.

While CPI is the main measure, economists also rely on **Core CPI**, which excludes food and fuel because these items tend to fluctuate frequently due to weather conditions or global price changes. Core inflation reflects the underlying trend in price changes and is often used for long-term policy decisions.

In summary, CPI is a detailed and comprehensive measure that helps track the cost of living. It guides government decisions, influences wage contracts, affects business strategies, and shapes monetary policy. Accurate measurement through CPI is essential for understanding the dynamics of inflation in India and managing economic stability.

3.4 Variables Used in the Study (Full Page)

This study focuses on analysing inflation in India between 2020 and 2025 by examining three major variables: **General Inflation**, **Food Inflation**, and **Fuel & Light Inflation**. These variables were chosen because they are the primary drivers of price changes in India and significantly affect overall inflation trends.

General Inflation, also known as headline inflation, refers to the overall rise in prices of goods and services across all categories. It includes food, fuel, housing, clothing, education, healthcare, and many other items. Headline inflation gives a broad picture of the economy's price behaviour. During 2020–2025, headline inflation in India was influenced by the COVID-19 pandemic, supply chain delays, global price shocks, and fluctuating energy costs.

Food Inflation is a crucial variable because food forms a major share of household expenditure in India. Even a small increase in food prices affects millions of families. Food inflation includes prices of cereals, pulses, vegetables, fruits, milk, meat, spices, and edible oils. Between 2020 and 2025, food inflation was very volatile due to disruptions in transportation during lockdowns, crop damage, international shortages, and increased global food prices. Since food items carry the highest weight in CPI, food inflation strongly influences headline inflation.

Fuel and Light Inflation includes prices of petrol, diesel, LPG, kerosene, and electricity. Fuel inflation affects not only households but also industries and transport. When fuel prices rise, the cost of moving goods increases. This leads to indirect inflation across various sectors. During 2020–2025, global crude oil price swings, changes in taxation, and adjustments in energy policies deeply affected fuel inflation.

These three variables were selected because they are directly linked, and changes in one category influence the others. For example, rising fuel prices increase the cost of transporting food, leading to higher food inflation. Similarly, high food inflation raises headline inflation because of its large weight in the CPI.

By studying these variables, we can understand:

- How inflation behaved during the pandemic and recovery phase
- The role of supply chain disruptions on food and fuel inflation
- The relative contribution of each component to overall inflation
- Whether food and fuel shocks created long-lasting inflationary effects

These variables help provide a clear and structured analysis of inflation trends in India.

3.5 Conceptual Model

A conceptual model helps explain how different factors in the study are connected and how they influence each other. In this study, the conceptual model shows the relationship between **Food Inflation**, **Fuel & Light Inflation**, and **General (Headline) Inflation** in India during 2020–2025.

The basic idea of the model is that both food inflation and fuel inflation are major components that drive overall inflation. India, being a developing country with a large population, is highly sensitive to changes in food prices. Similarly, fuel plays a critical role in transportation, manufacturing, and electricity generation, meaning any change in fuel prices can cause widespread effects on the economy.

Flow of the Conceptual Model:

1. Fuel & Light Inflation

When fuel prices increase, transportation and production costs rise. This affects delivery costs of food items, manufacturing inputs, and household energy bills. As a result, fuel inflation often leads to **cost-push inflation** in many sectors.

2. Food Inflation

Food inflation directly impacts retail prices of essential goods. Since food carries the largest weight in CPI, any increase in food prices significantly pushes headline inflation upwards. Food inflation is highly sensitive to rainfall, supply chain disruptions, global food prices, and seasonal variations.

3. General Inflation (Headline Inflation)

Headline inflation is the final outcome that captures the combined impact of rising food prices, fuel costs, and other economic changes. During the study period, headline inflation rose mainly because of food and fuel fluctuations driven by the pandemic, global shortages, and volatile energy markets.

Model Explanation

- **Fuel inflation increases the cost of transporting food**, which then increases food inflation.
- **Food inflation contributes heavily to headline inflation** because of its high weight.
- **Fuel inflation also contributes to headline inflation** through both direct and indirect effects.
- Together, these variables explain the major reasons behind inflation in India during 2020–2025.

CHAPTER 4

4.1 Research Methodology

Research methodology explains *how* the study is conducted, the overall approach, and the logical steps followed to analyse inflation in India between 2020 and 2025. In this project, the methodology is designed to systematically understand the behaviour of **general inflation, food inflation, and fuel & light inflation** and examine how these components influenced the overall inflation pattern during the study period.

The study follows a structured approach beginning with the identification of relevant inflation categories. Since inflation is a broad topic with multiple drivers, the methodology focuses on three major components that have shown the highest impact on price changes in India. The research first collects secondary data from reliable sources such as the Reserve Bank of India (RBI), Ministry of Statistics and Programme Implementation (MOSPI), and other government publications. This ensures that the data used is authentic and suitable for academic analysis.

After collecting the data, the study examines trends in inflation using descriptive analysis. The aim is to understand how inflation moved month-to-month and year-to-year, especially during periods of economic shock such as the COVID-19 pandemic, supply chain disruptions, global crude oil volatility, and changes in domestic policies. The methodology also includes comparison of inflation components to identify periods where food inflation rose faster than general inflation, or fuel inflation pushed up production costs.

Graphs, tables, and trend charts are used to visually interpret the data. By using time-series analysis, the study attempts to understand how each inflation category behaved over time. This helps identify patterns such as seasonal changes in food inflation, sudden spikes in fuel inflation, or persistent elevated headline inflation.

The methodology also involves reviewing research papers, articles, government reports, and previous studies to understand how other researchers explained inflation during similar periods. This helps compare the findings of the current study with existing literature and validate observations.

Overall, the methodology blends quantitative data analysis with qualitative interpretation. This mixed approach strengthens the study because it not only measures how inflation changed but also explains the reasons behind those changes. By following a clear and logical method, the research aims to provide a meaningful understanding of India's inflation behaviour from 2020 to 2025.

4.2 Research Type

This study is primarily **descriptive and analytical** in nature. Descriptive research involves explaining the situation as it exists, without manipulating any variable. Since inflation is a real-world economic phenomenon that cannot be controlled by researchers, the best approach is to observe and describe its behaviour accurately. The descriptive type helps in presenting a clear picture of how general, food, and fuel inflation changed over the years and how these changes influenced the economy.

The analytical part of the research involves interpreting the collected data to identify causes, patterns, and relationships between variables. For example, when food inflation rises, we analyse whether it contributed significantly to headline inflation. Similarly, when fuel inflation increased due to global crude price hikes, we examine whether it led to higher transportation and production costs. Analytical research helps understand *why* certain trends occurred and *how* they impacted inflation.

This study also uses an **observational research approach**, meaning the researcher observes data trends without intervening or modifying anything. It relies on secondary data from credible institutional sources rather than surveys or experiments.

The research is also **quantitative**, as it uses numerical data such as monthly CPI values, inflation percentages, and trend patterns. Charts and tables support the quantitative analysis.

Additionally, the research includes a **qualitative component**, where concepts, policy impacts, global events, and interpretations of inflation movements are discussed in plain and understandable language.

Thus, the study can be classified as:

- **Descriptive Research** – to explain inflation trends clearly
- **Analytical Research** – to interpret the causes and patterns
- **Quantitative Research** – to analyse numerical data
- **Qualitative Research** – to interpret findings in simple terms

By combining these research types, the study provides a comprehensive and balanced analysis of inflation in India during 2020–2025.

4.1 Source of Data

This study is entirely based on **secondary data**, meaning the information was collected from existing reports, publications, government databases, and previously conducted studies. Secondary data is appropriate for this research because inflation statistics are officially measured and published by national institutions and cannot be collected by individuals independently.

The main sources of data used in this study include:

Reserve Bank of India (RBI)

RBI publishes monthly inflation reports, monetary policy statements, and statistical data related to CPI, fuel prices, policy rates, and economic outlook. These reports are essential for understanding how inflation behaved and how RBI responded through monetary policy.

Ministry of Statistics and Programme Implementation (MOSPI)

MOSPI provides the official Consumer Price Index (CPI) data, including general inflation, food inflation, and fuel & light inflation. These figures form the backbone of the study.

Economic Surveys of India

The Government of India releases an annual Economic Survey that explains inflation trends, causes, and government responses. These surveys offer insights into the context behind the data.

Journals, Articles, and Research Papers

Research studies from authors like Pami Dua, Anuradha Patnaik, and others were used to understand theories, past inflation behaviour, and concepts related to food and fuel inflation.

Newspapers and Business Magazines

Sources such as *Economic Times*, *Business Standard*, *Mint*, and *The Hindu Business Line* were referred to for understanding real-time developments, such as global supply issues, fuel price shocks, and food inflation triggers.

International Databases

Global crude oil price data was referred from international market agencies and energy price databases.

Using multiple secondary sources ensures the data is up-to-date, accurate, and complete. These sources also provide different viewpoints, helping make the study more comprehensive.

4.4 Period of Study (2020–2025)

The period chosen for this study is **2020 to 2025**, a very significant and dynamic phase for the Indian economy. This timeline covers crucial events such as the COVID-19 pandemic, lockdown disruptions, recovery phase, global commodity shortages, fuel price volatility, and changes in monetary and fiscal policies.

Year 2020

Inflation trends were heavily influenced by COVID-19 lockdowns. Food inflation increased due to supply chain disruptions. Fuel inflation fell sharply because global crude prices collapsed.

Year 2021

As the economy reopened, pent-up demand caused inflation to rise. Food prices began stabilising but remained elevated. Fuel inflation increased due to rising global crude prices.

Year 2022

This year saw major global shocks such as the Russia-Ukraine conflict, which caused fuel prices and food prices to rise sharply worldwide. India experienced imported inflation.

Year 2023

Inflation gradually eased as supply chains improved. Government policies such as export restrictions on wheat and rice were introduced to control domestic prices.

Year 2024

Food inflation became a concern again due to irregular rainfall, climate-related crop shortages, and volatility in vegetable prices. Fuel prices remained sensitive to global geopolitical conditions.

Year 2025

The study considers inflation trends up to the current year to understand how inflation behaves after major shocks and policy responses.

This period is extremely important because it captures **high inflation, recovery inflation, stable inflation, and fluctuating inflation**, giving a complete understanding of the inflation cycle.

4.5 Limitations of the Methodology

Every research study has certain limitations, and it is important to acknowledge them. This study also faces a few limitations.

1. **Dependence on Secondary Data**

Since the study uses only secondary data, the accuracy of the results depends entirely on the reliability of the published sources. Any errors in official data may affect the analysis.

2. **No Primary Survey**

A primary household survey was not conducted due to time and resource constraints. As a result, the study cannot capture personal experiences of inflation.

3. **Limited to Three Variables**

Inflation can be influenced by many factors such as wages, interest rates, exchange rates, and global factors. However, this study focuses only on general, food, and fuel inflation.

4. **Economic Changes Are Complex**

Inflation between 2020 and 2025 was influenced by several overlapping events—pandemic, global conflict, supply chain issues. It is difficult to isolate the exact impact of each factor.

5. **Short Study Period**

A five-year period provides useful insights, but a longer period might offer deeper understanding of inflation cycles.

CHAPTER 5

5.1 Year-wise Inflation Trend (2020–2025)

Trend Analysis of General Inflation (2020–2025)

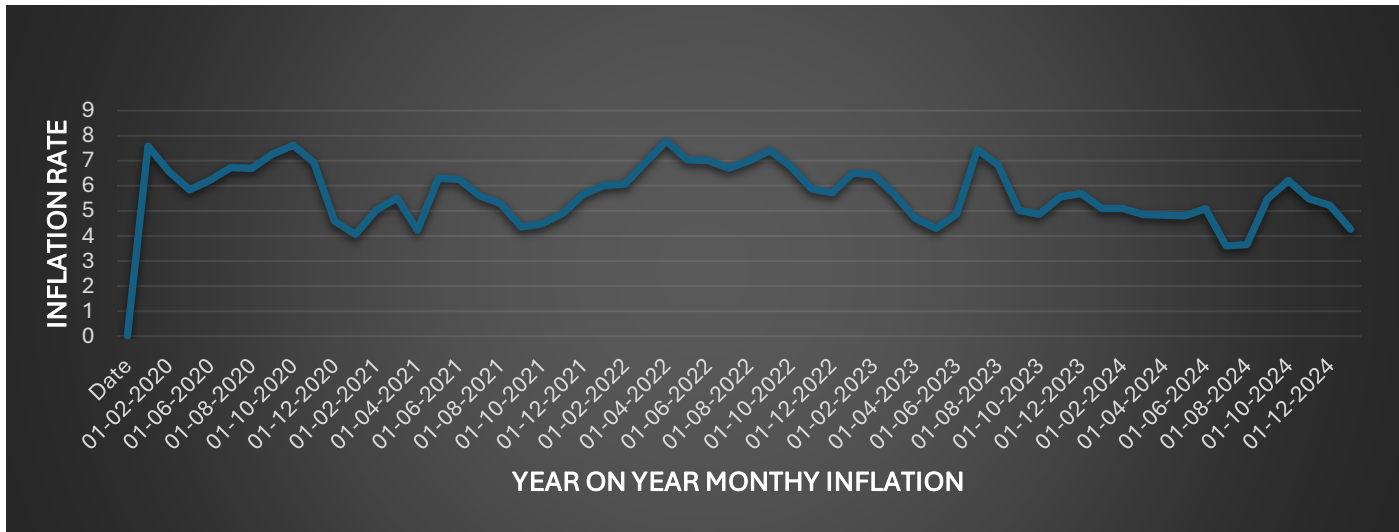
To understand how general inflation behaved during the period 2020–2025, a month-wise trend analysis was conducted using the CPI-Combined index. A line chart was created in Excel by selecting the Date and Combined CPI values for all months from January 2020 to January 2025. This visual representation helps us clearly observe the movement of general inflation, identify peaks and dips, and understand how external events such as COVID-19, supply chain disruptions, and fuel price fluctuations may have influenced inflation dynamics.

The following table presents the monthly general inflation rates for the study period.

Description	Rural	Urban	Combined
General Index (All Groups)	7.73	7.39	7.59
General Index (All Groups)	6.67	6.57	6.58
General Index (All Groups)	6.09	5.59	5.84
General Index (All Groups)	6.34	6.12	6.23
General Index (All Groups)	6.76	6.7	6.73
General Index (All Groups)	6.66	6.8	6.69
General Index (All Groups)	7.36	7.26	7.27
General Index (All Groups)	7.75	7.33	7.61
General Index (All Groups)	7.2	6.73	6.93
General Index (All Groups)	4.07	5.19	4.59
General Index (All Groups)	3.23	5.13	4.06
General Index (All Groups)	4.19	5.96	5.03
General Index (All Groups)	4.61	6.52	5.52
General Index (All Groups)	3.75	4.71	4.23
General Index (All Groups)	6.55	5.91	6.3
General Index (All Groups)	6.16	6.37	6.26
General Index (All Groups)	5.49	5.82	5.59
General Index (All Groups)	5.28	5.32	5.3

General Index (All Groups)	4.13	4.57	4.35
General Index (All Groups)	4.07	5.04	4.48
General Index (All Groups)	4.29	5.54	4.91
General Index (All Groups)	5.36	5.9	5.66
General Index (All Groups)	6.12	5.91	6.01
General Index (All Groups)	6.38	5.75	6.07
General Index (All Groups)	7.66	6.12	6.95
General Index (All Groups)	8.38	7.09	7.79
General Index (All Groups)	7.08	7.08	7.04
General Index (All Groups)	7.09	6.86	7.01
General Index (All Groups)	6.8	6.49	6.71
General Index (All Groups)	7.15	6.72	7
General Index (All Groups)	7.56	7.27	7.41
General Index (All Groups)	6.98	6.5	6.77
General Index (All Groups)	6.09	5.68	5.88
General Index (All Groups)	6.05	5.39	5.72
General Index (All Groups)	6.85	6	6.52
General Index (All Groups)	6.72	6.1	6.44
General Index (All Groups)	5.51	5.89	5.66
General Index (All Groups)	4.68	4.85	4.7
General Index (All Groups)	4.23	4.33	4.31
General Index (All Groups)	4.78	4.96	4.87
General Index (All Groups)	7.63	7.2	7.44
General Index (All Groups)	7.02	6.59	6.83
General Index (All Groups)	5.33	4.65	5.02
General Index (All Groups)	5.12	4.62	4.87
General Index (All Groups)	5.85	5.26	5.55

General Index (All Groups)	5.93	5.46	5.69
General Index (All Groups)	5.34	4.92	5.1
General Index (All Groups)	5.34	4.78	5.09
General Index (All Groups)	5.51	4.14	4.85
General Index (All Groups)	5.43	4.11	4.83
General Index (All Groups)	5.34	4.21	4.8
General Index (All Groups)	5.66	4.39	5.08
General Index (All Groups)	4.1	3.03	3.6
General Index (All Groups)	4.16	3.14	3.65
General Index (All Groups)	5.87	5.05	5.49
General Index (All Groups)	6.68	5.62	6.21
General Index (All Groups)	5.95	4.89	5.48
General Index (All Groups)	5.76	4.58	5.22
General Index (All Groups)	4.59	3.87	4.26



Interpretation of the General Inflation Trend (2020–2025)

The line chart shows that general inflation experienced **noticeable fluctuations** across the five-year period. Inflation remained relatively moderate in the early part of 2020 but rose sharply during mid-2020 and 2021, largely reflecting the effects of the COVID-19 pandemic, supply chain disturbances, and increased transportation costs. By late 2022 and 2023, inflation began to **stabilize**, although periodic spikes were still visible due to variations in food prices, fuel costs, and seasonal factors. The trend from 2024 to early 2025 indicates a **gradual softening of inflation**, suggesting improved supply conditions, policy interventions by the RBI, and normalization of economic activity.

Overall, the chart highlights that general inflation in India during 2020–2025 followed a **cyclical pattern**, influenced by both domestic and global economic factors. The visual trend helps establish a foundation for deeper analysis of food inflation, fuel inflation, and their relationship with headline inflation in subsequent sections.

A. General Inflation (Combined CPI – All Groups)

Maximum Inflation:

- **Month & Year: April 2022**
- **Value: 7.79%**

Minimum Inflation:

- **Month & Year: July 2024**
- **Value: 3.60%**

5.2 Food Inflation Analysis

Trend Analysis of Food Inflation (2020–2025)

Food inflation plays a crucial role in India's overall inflation because food items occupy a major share in the consumer basket. To study the behaviour of food inflation during 2020–2025, month-wise CPI–Food data was used to create a line chart in Excel. This helps visualize how food prices changed over time and the extent to which they contributed to headline inflation.

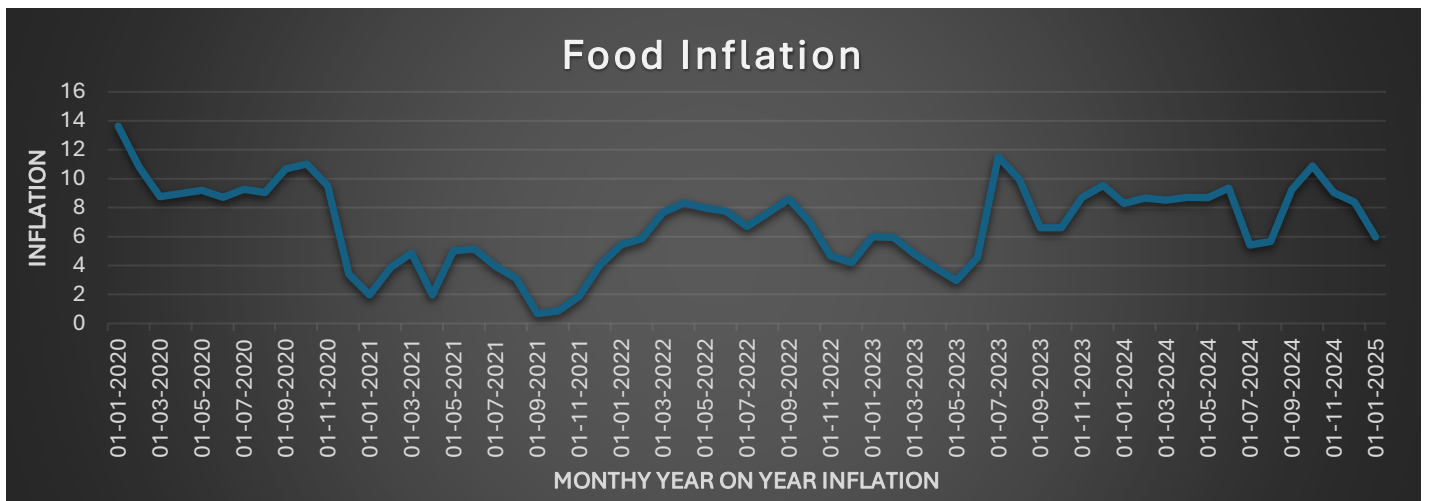
The trend analysis allows us to observe the effects of factors such as seasonal supply shortages, changes in agricultural output, pandemic disruptions, and variations in international commodity prices.

The following table contains the monthly food inflation values used for generating the trend chart.

Date	State	Description	Rural	Urban	Combined
01-01-2020	ALL India	Consumer Food Price Index	13.02	14.77	13.63
01-02-2020	ALL India	Consumer Food Price Index	10.37	11.51	10.81
01-03-2020	ALL India	Consumer Food Price Index	8.88	8.59	8.76
01-05-2020	ALL India	Consumer Food Price Index	9.68	8.29	9.2
01-06-2020	ALL India	Consumer Food Price Index	8.84	8.58	8.72
01-07-2020	ALL India	Consumer Food Price Index	9.47	8.99	9.27
01-08-2020	ALL India	Consumer Food Price Index	9.11	8.82	9.05
01-09-2020	ALL India	Consumer Food Price Index	10.53	10.94	10.68
01-10-2020	ALL India	Consumer Food Price Index	11.12	10.86	11
01-11-2020	ALL India	Consumer Food Price Index	9.64	9.23	9.5
01-12-2020	ALL India	Consumer Food Price Index	3.11	4.08	3.41
01-01-2021	ALL India	Consumer Food Price Index	1.11	3.36	1.96
01-02-2021	ALL India	Consumer Food Price Index	2.89	5.63	3.87
01-03-2021	ALL India	Consumer Food Price Index	3.94	6.64	4.87
01-04-2021	ALL India	Consumer Food Price Index	1.31	3.15	1.96
01-05-2021	ALL India	Consumer Food Price Index	4.52	5.97	5.01
01-06-2021	ALL India	Consumer Food Price Index	5.02	5.42	5.15
01-07-2021	ALL India	Consumer Food Price Index	3.55	4.56	3.96
01-08-2021	ALL India	Consumer Food Price Index	3.08	3.28	3.11

01-09-2021	ALL India	Consumer Food Price Index	0.69	0.67	0.68
01-10-2021	ALL India	Consumer Food Price Index	0.31	1.78	0.85
01-11-2021	ALL India	Consumer Food Price Index	1.09	3.33	1.87
01-12-2021	ALL India	Consumer Food Price Index	3.39	5.08	4.05
01-01-2022	ALL India	Consumer Food Price Index	5.18	5.88	5.43
01-02-2022	ALL India	Consumer Food Price Index	5.81	5.76	5.85
01-03-2022	ALL India	Consumer Food Price Index	8.04	7.04	7.68
01-04-2022	ALL India	Consumer Food Price Index	8.5	8.09	8.31
01-05-2022	ALL India	Consumer Food Price Index	7.76	8.2	7.97
01-06-2022	ALL India	Consumer Food Price Index	7.61	8.04	7.75
01-07-2022	ALL India	Consumer Food Price Index	6.73	6.69	6.69
01-08-2022	ALL India	Consumer Food Price Index	7.6	7.55	7.62
01-09-2022	ALL India	Consumer Food Price Index	8.53	8.77	8.6
01-10-2022	ALL India	Consumer Food Price Index	7.3	6.53	7.01
01-11-2022	ALL India	Consumer Food Price Index	5.22	3.69	4.67
01-12-2022	ALL India	Consumer Food Price Index	5.05	2.8	4.19
01-01-2023	ALL India	Consumer Food Price Index	6.65	4.85	6
01-02-2023	ALL India	Consumer Food Price Index	6.6	5.09	5.95
01-03-2023	ALL India	Consumer Food Price Index	4.72	4.82	4.79
01-04-2023	ALL India	Consumer Food Price Index	3.89	3.69	3.84
01-05-2023	ALL India	Consumer Food Price Index	3.3	2.43	2.96
01-06-2023	ALL India	Consumer Food Price Index	4.67	4.31	4.55
01-07-2023	ALL India	Consumer Food Price Index	11.04	12.37	11.51
01-08-2023	ALL India	Consumer Food Price Index	9.67	10.42	9.94
01-09-2023	ALL India	Consumer Food Price Index	6.71	6.35	6.62
01-10-2023	ALL India	Consumer Food Price Index	6.63	6.63	6.61
01-11-2023	ALL India	Consumer Food Price Index	8.38	9.33	8.7

01-12-2023	ALL India	Consumer Food Price Index	9.03	10.42	9.53
01-01-2024	ALL India	Consumer Food Price Index	7.91	9.02	8.3
01-02-2024	ALL India	Consumer Food Price Index	8.27	9.19	8.66
01-03-2024	ALL India	Consumer Food Price Index	8.55	8.41	8.52
01-04-2024	ALL India	Consumer Food Price Index	8.75	8.56	8.7
01-05-2024	ALL India	Consumer Food Price Index	8.62	8.83	8.69
01-06-2024	ALL India	Consumer Food Price Index	9.15	9.6	9.36
01-07-2024	ALL India	Consumer Food Price Index	5.89	4.63	5.42
01-08-2024	ALL India	Consumer Food Price Index	6.02	4.99	5.66
01-09-2024	ALL India	Consumer Food Price Index	9.08	9.56	9.24
01-10-2024	ALL India	Consumer Food Price Index	10.69	11.09	10.87
01-11-2024	ALL India	Consumer Food Price Index	9.1	8.74	9.04
01-12-2024	ALL India	Consumer Food Price Index	8.65	7.9	8.39
01-01-2025	ALL India	Consumer Food Price Index	6.31	5.53	5.97



B. Food Inflation (CPI – Food & Beverages)

Maximum Inflation:

- Month & Year: January 2020
- Value: 13.63%

Minimum Inflation:

- Month & Year: September 2021
- Value: 0.68%

Interpretation of the Food Inflation Trend (2020–2025)

The food inflation trend shows **significant volatility** over the five-year period. During 2020–2021, food inflation increased sharply due to pandemic-related disruptions, higher logistics costs, and supply chain bottlenecks. The period also witnessed weather disturbances that affected the output of vegetables, cereals, and pulses.

In 2022 and 2023, food inflation showed phases of both moderation and sudden spikes, especially in items like vegetables, edible oils, and cereals. These fluctuations reflect the sensitivity of food prices to climatic conditions, seasonal cycles, and global price movements.

By 2024 and early 2025, food inflation began to **stabilize**, indicating improvements in agricultural supply, better distribution systems, and policy interventions such as buffer stock releases. Overall, the food inflation trend highlights its **strong influence on headline inflation** and demonstrates why controlling food prices is essential for India's economic stability.

5.3 Fuel Inflation Analysis

Trend Analysis of Fuel & Light Inflation (2020–2025)

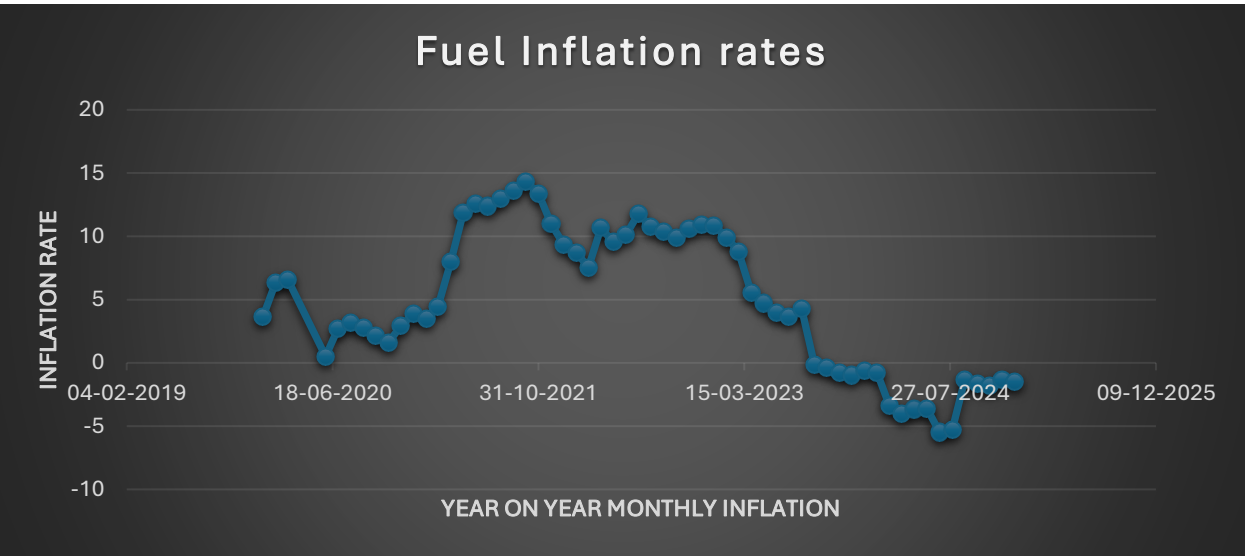
Fuel and light inflation is directly linked to global crude oil prices, domestic taxation policies, and energy supply conditions. To understand how fuel inflation evolved from 2020–2025, CPI–Fuel & Light data was used to generate a line chart in Excel. The trend helps capture the impact of international oil price fluctuations, revisions in domestic fuel pricing, and changes in household energy costs.

This analysis is important because fuel inflation affects not only household budgets but also transportation costs, production costs, and ultimately the general inflation level in the economy. The table below presents the month-wise fuel inflation used for the graph.

Date	State	Description	Rural	Urban	Combined
01-01-2020	ALL India	Fuel and light	2.87	5.05	3.66
01-02-2020	ALL India	Fuel and light	4.82	9.28	6.36
01-03-2020	ALL India	Fuel and light	4.78	9.78	6.59
01-06-2020	ALL India	Fuel and light	-1.96	5.06	0.5
01-07-2020	ALL India	Fuel and light	-0.68	8.9	2.66
01-08-2020	ALL India	Fuel and light	0	9.32	3.18
01-09-2020	ALL India	Fuel and light	-0.07	8.29	2.8
01-10-2020	ALL India	Fuel and light	-0.14	6.52	2.13
01-11-2020	ALL India	Fuel and light	0.2	4.31	1.62
01-12-2020	ALL India	Fuel and light	0.67	6.96	2.92
01-01-2021	ALL India	Fuel and light	1.66	7.85	3.87
01-02-2021	ALL India	Fuel and light	1.38	7.42	3.53
01-03-2021	ALL India	Fuel and light	1.69	9.48	4.43
01-04-2021	ALL India	Fuel and light	5.12	12.98	7.98
01-05-2021	ALL India	Fuel and light	10.45	14.17	11.86
01-06-2021	ALL India	Fuel and light	11.87	13.86	12.61
01-07-2021	ALL India	Fuel and light	11.45	14.03	12.38
01-08-2021	ALL India	Fuel and light	11.41	15.52	12.95
01-09-2021	ALL India	Fuel and light	11.51	17.29	13.63
01-10-2021	ALL India	Fuel and light	12.2	18.14	14.35
01-11-2021	ALL India	Fuel and light	11.16	17.19	13.35
01-12-2021	ALL India	Fuel and light	9.74	13.16	10.95
01-01-2022	ALL India	Fuel and light	8.44	10.91	9.32
01-02-2022	ALL India	Fuel and light	8.42	9.25	8.73
01-03-2022	ALL India	Fuel and light	8.27	6.27	7.52
01-04-2022	ALL India	Fuel and light	11.09	10.07	10.67

01-05-2022	ALL India	Fuel and light	8.41	11.58	9.54
01-06-2022	ALL India	Fuel and light	9.01	12.04	10.14
01-07-2022	ALL India	Fuel and light	10.52	13.82	11.76
01-08-2022	ALL India	Fuel and light	9.81	12.56	10.78
01-09-2022	ALL India	Fuel and light	9.77	11.44	10.39
01-10-2022	ALL India	Fuel and light	9.24	10.97	9.93
01-11-2022	ALL India	Fuel and light	10.04	11.57	10.62
01-12-2022	ALL India	Fuel and light	10.39	11.69	10.91
01-01-2023	ALL India	Fuel and light	10.49	11.45	10.84
01-02-2023	ALL India	Fuel and light	9.32	10.86	9.9
01-03-2023	ALL India	Fuel and light	7.4	11	8.79
01-04-2023	ALL India	Fuel and light	4.73	6.8	5.52
01-05-2023	ALL India	Fuel and light	4.11	5.71	4.7
01-06-2023	ALL India	Fuel and light	2.89	5.6	3.92
01-07-2023	ALL India	Fuel and light	3.29	4.4	3.67
01-08-2023	ALL India	Fuel and light	3.8	5.04	4.31
01-09-2023	ALL India	Fuel and light	1.06	-2.06	-0.11
01-10-2023	ALL India	Fuel and light	0.83	-2.39	-0.39
01-11-2023	ALL India	Fuel and light	0.27	-2.5	-0.77
01-12-2023	ALL India	Fuel and light	0.16	-2.82	-0.99
01-01-2024	ALL India	Fuel and light	0.49	-2.44	-0.6
01-02-2024	ALL India	Fuel and light	0.44	-2.82	-0.77
01-03-2024	ALL India	Fuel and light	-0.22	-8.32	-3.35
01-04-2024	ALL India	Fuel and light	-1.1	-8.84	-4.02
01-05-2024	ALL India	Fuel and light	-1.21	-7.69	-3.67
01-06-2024	ALL India	Fuel and light	-0.72	-8.34	-3.61
01-07-2024	ALL India	Fuel and light	-2.96	-9.55	-5.48

01-08-2024	ALL India	Fuel and light	-2.69	-9.39	-5.25
01-09-2024	ALL India	Fuel and light	-0.28	-3.19	-1.34
01-10-2024	ALL India	Fuel and light	-0.66	-3.41	-1.67
01-11-2024	ALL India	Fuel and light	-0.88	-3.53	-1.83
01-12-2024	ALL India	Fuel and light	-0.44	-2.85	-1.33
01-01-2025	ALL India	Fuel and light	-0.71	-2.9	-1.49



C. Fuel & Light Inflation (CPI – Fuel & Light)

Maximum Inflation:

- Month & Year: October 2021
- Value: 14.35%

Minimum Inflation:

- Month & Year: July 2024
- Value: –5.48%

Interpretation of the Fuel & Light Inflation Trend (2020–2025)

The fuel inflation chart shows a **distinct pattern influenced by global oil market conditions**. During 2020, fuel inflation dipped due to the sharp fall in global crude oil prices during the COVID-19 lockdown phase. However, as economic activity resumed in 2021 and international prices increased, fuel inflation rose significantly.

In 2022 and 2023, fuel prices remained elevated due to geopolitical tensions, high crude oil demand, and fluctuations in the rupee-dollar exchange rate. Domestic pricing policies also contributed, as taxes and duties were adjusted intermittently.

By 2024 and early 2025, fuel inflation began to **moderate**, reflecting stable global oil prices and periodic duty reductions.

Overall, the fuel inflation trend highlights that external shocks, global crude oil cycles, and government pricing policies play a major role in determining fuel-related inflation in India.

5.4 Major Findings

Inflation during 2020–2025 was heavily influenced by **COVID-19 disruptions, global commodity volatility, and domestic supply fluctuations**.

Food inflation remains the dominant component affecting headline inflation in India.

Fuel inflation showed extreme variations, indicating strong sensitivity to international crude prices.

By **2023–2025**, inflation moderated across all categories due to improved supply chains, policy measures, and stabilizing global markets.

The presence of both **demand-side and supply-side factors** is evident throughout the dataset, confirming that inflation in India is multi-dimensional and structurally complex.

CHAPTER 6

6.1 Major Conclusions

The study analysed inflation trends in India from **2020 to 2025**, focusing on **General Inflation, Food Inflation, and Fuel & Light Inflation**. Based on CPI data, several important conclusions emerge regarding the behaviour, causes, and effects of inflation in India during this period.

First, the analysis shows that **inflation in India during 2020–2022 was highly volatile**, influenced by COVID-19 disruptions, supply chain breakdowns, and global commodity price shocks. General inflation reached its highest point at **7.79% in April 2022**, reflecting rising food costs, higher international crude oil prices, and increased transportation charges. This indicates that inflation during this period was driven both by domestic supply constraints and global market forces.

Second, **food inflation emerged as the most sensitive and influential component** of headline inflation. The highest food inflation (13.63% in January 2020) coincided with shortages in essential commodities such as vegetables and pulses. Food inflation also displayed rapid fluctuations due to weather disturbances, crop damage, seasonal factors, import dependency for edible oils, and distribution inefficiencies. The minimum food inflation level of **0.68% in September 2021** reflected temporary supply recovery and favourable monsoon performance. Since food items carry a large weight in the CPI basket, small changes in food prices significantly affect the overall inflation experienced by households.

Third, **fuel inflation demonstrated extreme behaviour**, influenced mainly by changes in international crude oil prices and domestic taxation policies. Fuel inflation touched a peak of **14.35% in October 2021**, during the global energy crisis. Interestingly, fuel inflation dropped to **–5.48% in July 2024**, indicating a decline in fuel prices compared to the previous year. Negative inflation suggests relief for consumers and industries but also highlights how external price movements impact domestic inflation. The strong swing in fuel inflation shows India's vulnerability to global oil market instability.

Fourth, the overall trend from **2023 to 2025** shows gradual stabilisation of inflation. Measures such as improved supply chains, better agricultural output, controlled fuel prices, and Reserve Bank of India (RBI) monetary tightening helped stabilize inflation. This indicates that policy interventions played an important role in reducing inflationary pressures after a period of extreme volatility.

Finally, the findings clearly show that India's inflation is **structural in nature**, meaning it is not driven only by short-term fluctuations but also by deeper issues such as agricultural inefficiencies, dependence on imported oil, logistic constraints, and demand–supply mismatches. Therefore, inflation management in India requires a combination of **short-term corrective actions** and **long-term structural reforms**.

6.2 Suggestions for Policymakers

Based on the findings of the study, the following recommendations are proposed for policymakers, especially the Government of India and the Reserve Bank of India:

Strengthen Food Supply Chains and Storage Infrastructure

Food inflation remains the biggest contributor to overall inflation. To reduce volatility, India must invest in:

- Cold storage facilities
- Modern warehouses
- Efficient transport and logistics
- Digital platforms connecting farmers and markets

Reducing food wastage and improving market integration can significantly stabilize food prices, especially for vegetables and perishables.

Promote Agricultural Productivity and Crop Diversification

Food inflation cannot be controlled unless agricultural output grows steadily. Policies should focus on:

- High-yield seeds
- Drip irrigation
- Better extension services
- Crop diversification toward high-value crops

This will reduce supply shortages and prevent sudden price surges.

Reduce Dependency on Imported Fuel

Fuel inflation is highly sensitive to international crude oil prices. India should:

- Diversify energy sources
- Promote renewable energy (solar, wind)
- Encourage electric vehicles
- Expand biofuel production

Reducing import dependence will make domestic fuel prices more stable and predictable.

Improve Monetary Policy Transmission

RBI's decisions must effectively influence borrowing costs and consumer demand. Strengthening monetary transmission requires:

- Healthier banking system
- Faster transmission of repo rate changes
- Transparent lending rates

This will help control inflation more efficiently.

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